S1 — The Vorton-Scalar, its neutral current, and the hidden-axis singlet superfluid of the 16-component Dirac field

Supplement to both manuscripts of PROJECT spacetimes: invention of the field, Lagrangians, field equations, pairing interaction, gap equation, order parameter — for the 7+1 and the 4+4 spacetime, with the provenance of every statement

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0. Purpose, and how to read this supplement

The task of the project is to *invent, characterize and describe by field equations* a superfluid state of spin 0 that is built only from a 16-component fermion field on an eight-dimensional flat spacetime, in which the fermions are paired by the exchange of a new scalar boson, the **Vorton-Scalar**, through a neutral current made only of that scalar. This supplement does that, for both spacetimes of the project:

- the spacetime of signature 7+1 (seven space dimensions, one time), field **dirac1671**;
- the spacetime of signature 4+4 (four space and four time dimensions), field **dirac1644**.

It assumes that the reader has the provenance **"dirac matrices"** at hand. That document defines the two sets of real 16×16 Dirac matrices, proves their properties, states the two Dirac equations, and shows which fermion pairs exist (its Part 3). Nothing proved there is proved again here; it is cited as "DM, Sec. …" or by the identifier of a machine check (for example `71-K3`).

The supplement is written for a physics graduate student who has not seen the solution. Every symbol is defined in words where it first appears; Sec. 0.2 collects them. Each section ends by saying what the next one needs. Section 9 separates what has been established from what has not.

0.1 The coordinates, once more

In both spacetimes the coordinates are $x0, x1, \dots, x7$, with the meaning fixed by the author's notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb` (DM, Sec. 0.1): $x1, x2, x3$ are the ordinary space coordinates; $x4$ is the ordinary time; $x0$ is a **hidden space-like coordinate**; $x5, x6, x7$ are three further coordinates, time-like in 4+4 and space-like in 7+1. Capital indices $A, B = 0, \dots, 7$ label all eight coordinates. Three smaller index sets are used throughout:

Index	Values	Meaning
i, j	1, 2, 3	ordinary space
a, b	0, 5, 6, 7	the hidden axis and the three further coordinates ("the four extra axes")
t	5, 6, 7	the three further coordinates alone

Natural units $\hbar = c = k_B = 1$ are used.

0.2 Symbols introduced in this supplement

The symbols of "dirac matrices" keep their meaning and their tags (1644 for 4+4, 1671 for 7+1). The symbols below are new. Their names in the files carry the suffix VS or a tag, which no symbol of the author's notebooks contains (MASTER provenance, `provenance/P1_master_symbols`, check `S6`). A few of the letters used in formulas carry neither (m_s, J_5, F_S, F_P, Δ^a, `M`); where such a letter also occurs in a notebook with another meaning, the MASTER provenance says so (its Sec. 6.3: m_s, for example, is the mass of the strange quark in one section of `7+1.nb`).

Symbol	Name in files	Meaning
Φ_{VS}	`PhiVS`	the Vorton-Scalar: a complex scalar field
v_{VS}	`vVS`	its vacuum expectation value, $\langle\Phi_{VS}\rangle = v_{VS}/\sqrt{2}$
λ_{VS}, y_{VS}	`lambdaVS`, `yVS`	self-coupling of Φ_{VS}; Yukawa coupling of Φ_{VS} to the fermion field
s_{VS}, m_s	`sVS`, `msVS`	radial mode of Φ_{VS} and its mass
ϑ_{VS}	`thetaVS`	phase mode of Φ_{VS} (massless)
g_{VS}	`gVS`	coupling of both modes to the fermions, $g_{VS} = m/v_{VS}$
α_{VS}	`alphaVS`	$g_{VS}^2/4\pi$
D_{VS}	`DVS`	$\alpha_{VS}\, m/m_s$, the number that decides whether two fermions bind
J^A_{VS}	`JVS^A`	the neutral current of the Vorton-Scalar, Eq. (1.6)
J^A_5	`J5^A`	the axial current of the fermion field, Eq. (1.6)
G_{VS}	`GVS`	the Fermi-type constant of the neutral-current interaction, $G_{VS}/\sqrt{2} = 1/(2v_{VS}^2)$
$U(1)_{VS}$, $U(1)_F$	—	the chiral symmetry carried by Φ_{VS}; fermion number
β, α^j, χ	`beta…`, `alpha…`, `chi…`	Hamiltonian matrices and chirality matrix of the quantized field, Eq. (2.1), with the tag of the spacetime
F_S, F_P	`F_scalar`, `F_pseudoscalar`	exchange amplitudes of the radial and of the phase mode in a pair channel, Eq. (4.2)
Δ^a	`DeltaV…^a`	order parameter: expectation value of the vector pair operator along the extra axis a
`M`	—	the 8×8 gap matrix, Eq. (6.1)

As in "dirac matrices", the tag is dropped inside a section that deals with one spacetime only, and boxed results carry all tags.

0.3 How to reproduce the numbers

```
cd "/Users/nsh/Developer/github/spacetimes"
cargo test --release --manifest-path code/spacetimes-rs/Cargo.toml     # 63 unit tests
cargo run  --release --manifest-path code/spacetimes-rs/Cargo.toml -- channels
cargo run  --release --manifest-path code/spacetimes-rs/Cargo.toml -- symmetry
cargo run  --release --manifest-path code/spacetimes-rs/Cargo.toml -- gap
cargo run  --release --manifest-path code/spacetimes-rs/Cargo.toml -- dimer
python3 code/scripts/make_figures.py kernels gap dimer
```

The programs are described in `provenance/P8_code`; the table in Sec. 10 says which command produces which number of this supplement.

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1. The invention: the Vorton-Scalar in 7+1

This section uses the 7+1 spacetime and drops the tag 1671. The 7+1 case comes first because it has one time and therefore a conventional quantum theory; Sec. 3 then says what changes in 4+4.

1.1 What is wanted

The free Lagrangian of the field (DM, Sec. 2.8.2) is $\mathcal{L}_0 = \bar\psi(i\gamma^A\partial_A - m)\psi$ with $\bar\psi = \psi^\dagger\gamma^4$. Pairing needs an attraction between two fermions. The task prescribes its origin: a *scalar* boson field, and an interaction of the kind known from the weak force, in which a **neutral current** — a current that does not change the kind of fermion it acts on — of one sector couples to a current of the fermions. The neutral current is to be built from the scalar alone.

A real scalar field has no current. The simplest field that has one is a *complex* scalar Φ_{VS}, whose phase rotations $\Phi_{VS} \to e^{i\epsilon}\Phi_{VS}$ are a symmetry; the conserved current of that symmetry is bilinear in Φ_{VS} and is the neutral current sought. For it to act on the fermions, the fermions must carry the same symmetry. A field with a mass term has only one phase symmetry, fermion number; but the massless field has a second one, the chiral rotation (DM, Sec. 2.6), and the mass term breaks it. This suggests the construction: **let the mass of the fermion field come from the Vorton-Scalar**, so that the chiral rotation of the fermions, accompanied by a phase rotation of Φ_{VS}, is an exact symmetry.

1.2 Definition

$$\boxed{\begin{aligned}\mathcal{L}^{VS}_{1671} = {}& \bar\psi\, i\gamma^A\partial_A\psi + \eta^{AB}\,\partial_A\Phi^*_{VS}\,\partial_B\Phi_{VS} - \lambda_{VS}\Big(|\Phi_{VS}|^2 - \frac{v_{VS}^2}{2}\Big)^2 \\ & - y_{VS}\big(\Phi_{VS}\,\bar\psi P^R\psi + \Phi^*_{VS}\,\bar\psi P^L\psi\big)\end{aligned}} \tag{1.1}$$

Here $\psi = \psi_{1671}$, $\gamma^A = \gamma^A_{1671}$, $\eta = \eta_{1671}$, and $P^{L,R} = \frac{1}{2}(1 \mp \chi)$ are the chiral projectors of 7+1 with the chirality matrix $\chi = \chi_{1671} = i\omega_{1671}$ (DM, Eq. (2.4)). The constants $\lambda_{VS} > 0$, $v_{VS} > 0$ and $y_{VS} > 0$ are real. The last term is Hermitian: $(\bar\psi P^R\psi)^\dagger = \psi^\dagger P^R\gamma^4\psi = \bar\psi P^L\psi$, because χ anticommutes with γ^4 (check `71-P1`).

Symmetries. Besides the Lorentz group of 7+1, the Lagrangian (1.1) has two phase symmetries:

$$U(1)_F:\ \psi \to e^{i\epsilon}\psi; \qquad U(1)_{VS}:\ \psi \to e^{i\epsilon\chi}\psi, \quad \Phi_{VS} \to e^{-2i\epsilon}\Phi_{VS}. \tag{1.2}$$

Proof for $U(1)_{VS}$. Since χ is Hermitian and anticommutes with every γ^A, $\bar\psi \to \bar\psi e^{i\epsilon\chi}$ and $\bar\psi\gamma^A\partial_A\psi \to \bar\psi e^{i\epsilon\chi}\gamma^A e^{i\epsilon\chi}\partial_A\psi = \bar\psi\gamma^A\partial_A\psi$; and $\bar\psi P^R\psi \to \bar\psi e^{2i\epsilon\chi}P^R\psi = e^{2i\epsilon}\bar\psi P^R\psi$, which the phase of Φ_{VS} compensates. A bare mass term $m\bar\psi\psi$ is *not* invariant and is therefore absent from (1.1). ■

The name. In the author's NCSS project the name Vorton-Scalar denotes the scalar that mediates the pairing and that is carried along by every quantized vortex of the superfluid; that meaning is kept here (the condensate vanishes in the core of a vortex, and by the second of Eqs. (1.8) the scalar field, whose source the fermions are, changes there with it). The field invented here is, in addition, *complex* with a broken phase symmetry. Such a field has line-like defects of its own, on which the fermions are massless and can carry a persistent current, and closed loops stabilized by such a current are what the literature on cosmic strings calls *vortons*. Section 8 keeps the two meanings apart.

1.3 Dimensions of the couplings in eight dimensions

In natural units the action $\int d^8x\,\mathcal{L}$ is dimensionless, so $[\mathcal{L}] = \text{mass}^8$. The kinetic terms give $[\psi] = \text{mass}^{7/2}$ and $[\Phi_{VS}] = \text{mass}^3$, hence

$$[v_{VS}] = \text{mass}^3, \qquad [y_{VS}] = \text{mass}^{-2}, \qquad [\lambda_{VS}] = \text{mass}^{-4}.$$

Couplings of negative mass dimension mean that (1.1) is an *effective* theory, valid below some scale; this is unavoidable for interacting fields in eight dimensions and is not specific to the present model.

In Stage 1 of the author's programme the fields that matter do not depend on the four extra coordinates (the hidden axis is curled up; the three others are assumed to play no dynamical role). If the extra coordinates span a volume V_4, a field that is constant along them is $\psi = \psi^{(4)}/\sqrt{V_4}$, $\Phi_{VS} = \Phi^{(4)}_{VS}/\sqrt{V_4}$, and the action becomes a four-dimensional one with the dimensionless couplings and the scale

$$y^{(4)}_{VS} = \frac{y_{VS}}{\sqrt{V_4}}, \qquad \lambda^{(4)}_{VS} = \frac{\lambda_{VS}}{V_4}, \qquad v^{(4)}_{VS} = v_{VS}\sqrt{V_4} \quad \text{(a mass)}.$$

From Sec. 2 on, all couplings are these four-dimensional ones and the superscript (4) is dropped.

1.4 Symmetry breaking: the mass of the fermions and the two modes

The potential in (1.1) is minimal at $|\Phi_{VS}| = v_{VS}/\sqrt{2}$. Write

$$\Phi_{VS}(x) = \frac{v_{VS} + s_{VS}(x)}{\sqrt{2}}\, e^{i\vartheta_{VS}(x)/v_{VS}} \tag{1.3}$$

with two real fields: the **radial mode** s_{VS} and the **phase mode** ϑ_{VS}. Because $\Phi P^R + \Phi^* P^L = |\Phi|\, e^{i\chi \arg\Phi}$, the Yukawa term becomes $-m\,(1 + s_{VS}/v_{VS})\,\bar\psi\, e^{i\chi\vartheta_{VS}/v_{VS}}\psi$, and (1.1) reads

$$\mathcal{L}^{VS}_{1671} = \bar\psi(i\gamma^A\partial_A - m)\psi + \tfrac{1}{2}(\partial s_{VS})^2 - \tfrac{1}{2}m_s^2 s_{VS}^2 + \tfrac{1}{2}(\partial\vartheta_{VS})^2 - g_{VS}\, s_{VS}\,\bar\psi\psi - g_{VS}\,\vartheta_{VS}\,\bar\psi\, i\chi\,\psi + \ldots \tag{1.4}$$

with

$$\boxed{m_{1671} = \frac{y_{VS}\, v_{VS}}{\sqrt{2}}, \qquad m_s = \sqrt{2\lambda_{VS}}\, v_{VS}, \qquad g_{VS} = \frac{m_{1671}}{v_{VS}} = \frac{y_{VS}}{\sqrt{2}}} \tag{1.5}$$

and the dots standing for terms with three or more boson fields. Three facts follow.

1. The fermion mass is no longer a free parameter put in by hand: it is the Yukawa coupling times the vacuum value of the Vorton-Scalar.
2. The radial mode is a scalar of mass m_s that couples to $\bar\psi\psi$.
3. The phase mode is massless (it is the Goldstone boson of the broken $U(1)_{VS}$) and couples to the pseudoscalar density $\bar\psi i\chi\psi$. Both couplings have the same strength $g_{VS} = m/v_{VS}$: heavier fermions couple more strongly, a relation of the Goldberger–Treiman type.

1.5 The neutral current

The conserved current of $U(1)_{VS}$ follows from (1.2) by Noether's theorem:

$$\partial_A(J^A_5 - 2J^A_{VS}) = 0, \qquad \boxed{J^A_{VS} = i(\Phi^*_{VS}\,\partial^A\Phi_{VS} - \Phi_{VS}\,\partial^A\Phi^*_{VS}), \qquad J^A_5 = \bar\psi\gamma^A\chi\psi} \tag{1.6}$$

J^A_{VS} is **the neutral current of the Vorton-Scalar**: it is built from Φ_{VS} alone, and it is neutral in the sense of the weak interaction, since it leaves the fermion that it acts on unchanged. J^A_5 is the axial current of the fermions. Neither is conserved separately: the Vorton-Scalar current supplies the divergence of the axial current.

The coupling of the phase mode in (1.4) can be written with these currents. Redefine the fermion field by a local chiral rotation, $\psi \to e^{-i\chi\vartheta_{VS}/2v_{VS}}\psi$. This removes ϑ_{VS} from the Yukawa term, and the kinetic term produces instead $\frac{1}{2v_{VS}}\,\partial_A\vartheta_{VS}\,J^A_5$. With (1.3), $J^A_{VS} = -(v_{VS} + s_{VS})^2\,\partial^A\vartheta_{VS}/v_{VS}$, so to lowest order in the fields $\partial^A\vartheta_{VS} = -J^A_{VS}/v_{VS}$ and

$$\boxed{\mathcal{L}_{\text{NC}} = -\frac{G_{VS}}{\sqrt{2}}\; J_{VS\,A}\, J^A_5, \qquad \frac{G_{VS}}{\sqrt{2}} = \frac{1}{2\,v^2_{VS}}} \tag{1.7}$$

This is a **current–current interaction of the weak type**, with a "Fermi constant" G_{VS} fixed by the vacuum value alone — just as the Fermi constant of the weak interaction is $G_F/\sqrt{2} = 1/(2v_{\rm EW}^2)$ in terms of the electroweak vacuum value. Equations (1.4) and (1.7) describe the same physics in two sets of field variables; (1.4) is used for the calculations below, (1.7) shows that the exchange of the phase mode *is* the neutral-current interaction that the task asks for.

What Sec. 2 needs from this section: the vertices of Eq. (1.4).

1.6 Field equations

Varying (1.1) with respect to $\bar\psi$ and Φ^*_{VS} gives the coupled field equations of the 7+1 model:

$$\boxed{\begin{aligned}&[i\gamma^A_{1671}\partial_A - y_{VS}(\Phi_{VS}P^R_{1671} + \Phi^*_{VS}P^L_{1671})]\psi_{1671} = 0,\\ &\eta^{AB}_{1671}\partial_A\partial_B\Phi_{VS} + 2\lambda_{VS}\Big(|\Phi_{VS}|^2 - \frac{v^2_{VS}}{2}\Big)\Phi_{VS} + y_{VS}\,\bar\psi_{1671}P^L_{1671}\psi_{1671} = 0.\end{aligned}} \tag{1.8}$$

For $\Phi_{VS} = v_{VS}/\sqrt{2}$ the first equation is the Dirac equation (2.9) of "dirac matrices" with the mass (1.5). The second shows that a fermion condensate $\langle\bar\psi P^L\psi\rangle \neq 0$ acts as a source of the Vorton-Scalar.

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2. The quantized theories in a common form

The pairing calculation uses the Hamiltonian form of the two Dirac equations. It is the same for both spacetimes:

$$i\,\partial_4\psi = \mathsf{h}\,\psi, \qquad \mathsf{h} = -i\,\alpha^j\partial_j + \beta\,m, \tag{2.1}$$

with Hermitian matrices β, α^j that satisfy $\beta^2 = (\alpha^j)^2 = 1$ and anticommute with each other, and a Hermitian chirality matrix χ, $\chi^2 = 1$, that commutes with the α^j and anticommutes with β:

	7+1 (field dirac1671)	4+4 (field dirac1644, independent of $x5, x6, x7$)
β	$\beta_{1671} = \gamma^4_{1671}$ (real symmetric)	$\beta_{1644} = -i\gamma^4_{1644}$ (imaginary antisymmetric)
α^j	$\gamma^4_{1671}\gamma^j_{1671}$, seven space indices j	$-\gamma^4_{1644}\gamma^j_{1644}$, four space indices $j = 0, 1, 2, 3$
chirality χ	$\chi_{1671} = i\omega_{1671}$ (imaginary)	ω_{1644} (real, diagonal)
source	DM, Eq. (2.11); checks `71-D4`, `71-D5`	DM, Eq. (1.17); checks `44-D6` to `44-D8`

In both cases $\mathsf{h}(\vec p)^2 = (|\vec p|^2 + m^2)\,1$: at each momentum there are eight states of energy $+E$ (particles) and eight of energy $-E$, whose holes are the antiparticles. The vertices of the two modes of the Vorton-Scalar are, from (1.4),

$$\text{radial mode: } -g_{VS}\,s_{VS}\,\psi^\dagger\beta\,\psi, \qquad \text{phase mode: } -g_{VS}\,\vartheta_{VS}\,\psi^\dagger(i\beta\chi)\psi. \tag{2.2}$$

The matrices β and $i\beta\chi$ are Hermitian.

2.1 The internal symmetry when the fields do not depend on the extra coordinates

The pairing calculation of Secs. 4 to 6 uses fields that do not depend on the four extra coordinates $x0, x5, x6, x7$: all momenta lie in ordinary space. In this **zero-mode sector** the dynamics is fixed by five Hermitian matrices only — $\alpha^1, \alpha^2, \alpha^3$, β and $i\beta\chi$ — and an *internal symmetry* is a unitary

transformation $\psi \to e^{i\theta \mathsf{g}}\psi$ of the sixteen components whose generator, a Hermitian matrix g, commutes with all five. The program `spacetimes symmetry` (`symmetry.rs`; `data/reduced_symmetry.csv`; unit test `symmetry::tests::zero_mode_symmetry_is_u2_times_u2_in_both_signatures`) computes the set of all such generators from the matrices of each spacetime. The result is **the same for 7+1 and 4+4**:

Property (computed)	7+1	4+4
number of independent generators, momenta along $x1, x2, x3$	8	8
structure of the algebra	$su(2)_A \oplus su(2)_B \oplus u(1)_F \oplus u(1)_E$	the same
the eight states of a particle of given momentum	$(2,1) \oplus (1,2)$	the same
the four vector pair operators $C^-\gamma^a$, $a \in \{0,5,6,7\}$	$(2,2)$, fermion number 2	the same
invariant quadratic form on these four operators	diag$(+,+,+,+)$	diag$(+,-,-,-)$
generators if the fields may also depend on $x0$	4: $su(2) \oplus u(1)_F$	4: $su(2)_T \oplus u(1)_F$

Here $u(1)_F$ is fermion number, and $u(1)_E$ is generated by a matrix P with $P^2 = 1$ that distinguishes the doublets of $SU(2)_A$ ($P = +1$) from those of $SU(2)_B$ ($P = -1$). The group $[SU(2)_A \times SU(2)_B]/Z_2$ is $SO(4)$, and the four pair operators form its 4-vector. The two spacetimes differ in *what the six generators are*:

- **In 7+1** they are $i\gamma^a\gamma^b$ with $a, b \in \{0,5,6,7\}$: the rotations of the four extra space axes, a symmetry of the full theory, in which they rotate the coordinates as well.
- **In 4+4** three of them are the rotations $i\gamma^s\gamma^t$ of the three additional times, and the other three are the matrices $\gamma^0\gamma^t$ ($t = 5,6,7$). In 4+4 these matrices are *Hermitian*, so that $e^{i\theta\gamma^0\gamma^t}$ is unitary. It is **not** the Lorentz boost $e^{\theta\gamma^0\gamma^t/2}$ that mixes $x0$ with a time coordinate (a boost is not unitary): the algebra of the boosts and the algebra found here are two different real forms, $so(1,3)$ and $so(4)$, of the same complex algebra. The $SO(4)$ of the 4+4 model is therefore a **hidden symmetry**: it acts on the components of the field, has no counterpart among the transformations of the coordinates, and exists only as long as the fields do not depend on $x0$ (last line of the table).

The invariant quadratic form in the last-but-one line is the metric of each spacetime on the four extra axes, up to an overall sign. It reappears in the energy of the superfluid, Eq. (6.2).

What Sec. 3 does: it justifies the right-hand column of the first table, and states exactly what in 4+4 is derived and what is assumed.

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3. The Vorton-Scalar in 4+4: what carries over, and what does not

This section uses the 4+4 spacetime and drops the tag 1644.

3.1 The covariant Lagrangian has no chiral phase symmetry

The Lagrangian of dirac1644 that is invariant under the whole group $\mathrm{Spin}_0(4,4)$ is $\mathcal{L} = \tilde{\psi}(\gamma^A\partial_A - m)\psi$ with $\tilde{\psi} = \psi^\dagger C^-$ (DM, Eq. (1.14)). In 4+4 the chirality matrix ω is real, and C^- — a product of four Dirac matrices — *commutes* with it. Therefore, under the chiral phase rotation $\psi \to e^{i\epsilon\omega}\psi$,

$$\tilde{\psi}\psi \to \tilde{\psi}\psi, \qquad \tilde{\psi}\gamma^A\partial_A\psi \to \tilde{\psi}\, e^{-2i\epsilon\omega}\gamma^A\partial_A\psi :$$

the *mass term* is invariant and the *kinetic term* is not — the opposite of the situation in 7+1 (and in ordinary four-dimensional theory). The kinetic term is invariant instead under the real, non-compact rescaling $\psi \to e^{\epsilon\omega}\psi$, which multiplies the two 8-component chiral parts by $e^{-\epsilon}$ and $e^{+\epsilon}$ and mixes $\tilde{\psi}\psi$ with $\tilde{\psi}\omega\psi$ by a hyperbolic rotation. A scalar that compensated this rotation would have the invariant $(\mathrm{Re}\,\Phi)^2 - (\mathrm{Im}\,\Phi)^2$ of indefinite sign: one of its two components would have negative kinetic energy. **In the fully covariant 4+4 theory the construction of Sec. 1 does not exist.** This is a consequence of the real chirality of 4+4 (DM, Theorem 1 and Sec. 1.6) and is recorded as a result, not as a difficulty to be hidden.

3.2 The quantized sector, where the construction does exist

"dirac matrices" (Sec. 1.8.4) shows that the field dirac1644 has a unitary evolution in the ordinary time $x4$ and a positive conserved density only on fields that do not depend on the additional times $x5, x6, x7$, and defines the quantized field of Stage 1 by that restriction, with the positive Lagrangian $\mathcal{L}^{\mathrm{red}} = \psi^\dagger(i\partial_4 - \mathsf{h})\psi$. It differs from the covariant Lagrangian by the matrix $-W$, $W = iC^-\gamma^4$, and W *anticommutes* with ω (it contains an odd number of Dirac matrices). In $\mathcal{L}^{\mathrm{red}}$ the roles are therefore the familiar ones: with $\beta = -i\gamma^4$ and $\alpha^j = -\gamma^4\gamma^j$,

$$\mathcal{L}^{\mathrm{red}} = \psi^\dagger(i\partial_4 + i\alpha^j\partial_j)\psi - m\,\psi^\dagger\beta\,\psi,$$

the first term is invariant under $\psi \to e^{i\epsilon\omega}\psi$ (ω commutes with α^j) and the mass term is not (ω anticommutes with β). The Vorton-Scalar of the 4+4 model is defined in this sector:

$$\begin{aligned}\mathcal{L}^{VS}_{1644} = {}& \psi^\dagger(i\partial_4 + i\alpha^j_{1644}\partial_j)\psi + |\partial_4\Phi_{VS}|^2 - \sum_{j=0}^{3}|\partial_j\Phi_{VS}|^2 - \lambda_{VS}\Big(|\Phi_{VS}|^2 - \frac{v_{VS}^2}{2}\Big)^2 \\ & - y_{VS}\,\psi^\dagger\beta_{1644}(\Phi_{VS}P^R_{1644} + \Phi^*_{VS}P^L_{1644})\psi\end{aligned} \tag{3.1}$$

with $\psi = \psi_{1644}$ independent of $x5, x6, x7$, j running over the four space coordinates $x0, \dots, x3$, and the real projectors $P^{L,R}_{1644} = \frac{1}{2}(1 \mp \omega_{1644})$. It is invariant under $\mathrm{Spin}(4,1) \times \mathrm{Spin}(3)$ — rotations and boosts of $(x0, x1, x2, x3; x4)$, and rotations of the three additional times — and under

$$U(1)_F:\ \psi \to e^{i\epsilon}\psi; \qquad U(1)_{VS}:\ \psi \to e^{i\epsilon\omega_{1644}}\psi, \quad \Phi_{VS} \to e^{-2i\epsilon}\Phi_{VS}.$$

Everything in Secs. 1.4 and 1.5 then holds with χ replaced by ω_{1644} and $\bar{\psi}$ by $\psi^\dagger\beta_{1644}$: $m_{1644} = y_{VS}v_{VS}/\sqrt{2}$, $g_{VS} = m_{1644}/v_{VS}$, the vertices (2.2), and the neutral current. The field equations are

$$\begin{aligned}&[i\partial_4 + i\alpha^j_{1644}\partial_j - y_{VS}\,\beta_{1644}(\Phi_{VS}P^R_{1644} + \Phi^*_{VS}P^L_{1644})]\psi_{1644} = 0, \\ &(\partial_4^2 - \textstyle\sum_{j=0}^3\partial_j^2)\Phi_{VS} + 2\lambda_{VS}\Big(|\Phi_{VS}|^2 - \frac{v_{VS}^2}{2}\Big)\Phi_{VS} + y_{VS}\,\psi^\dagger_{1644}\beta_{1644}P^L_{1644}\psi_{1644} = 0.\end{aligned} \tag{3.2}$$

What is assumed. The restriction to fields independent of $x5, x6, x7$ is the author's hypothesis that the additional times deflate away; in flat spacetime it is imposed, not derived. The model (3.1) is a consistent quantum theory of a 16-component field in four space dimensions and one time; whether it is the low-energy form of a theory on the full 4+4 spacetime with gravity is Stage 2 of the programme and is not examined here.

What Sec. 4 needs: only Eqs. (2.1) and (2.2), which now hold for both spacetimes.

4. The pairing interaction

4.1 Definition of the exchange amplitude in a pair channel

A superfluid forms in a medium of fermions. Its rest frame is the frame of the medium, and the momenta of the fermions lie in ordinary space, $\vec{p} = (p^1, p^2, p^3)$ (the fields do not depend on the extra coordinates). Let $U(\vec{p})$ be the 16×8 matrix whose columns are orthonormal positive-energy spinors, $\mathsf{h}(\vec{p})U = EU$, so that the particle part of the field is $\psi(\vec{x}) \supset \sum_{\vec{p}} U(\vec{p})\, b(\vec{p})\, e^{i\vec{p}\cdot\vec{x}}$ with a column $b(\vec{p})$ of eight annihilation operators.

A *pair operator* $\psi^T X \psi$ (DM, Part 3; X a 16×16 matrix) contains the product $b_r(\vec{p})\, b_s(-\vec{p})$ with the coefficient

$$\Phi_X(\vec{p}) = U(\vec{p})^T\, X\, U(-\vec{p}) \qquad (8 \times 8), \tag{4.1}$$

its **two-particle content**. One boson exchanged between the two fermions scatters the pair $(\vec{p}, -\vec{p})$ into $(\vec{p}\,', -\vec{p}\,')$. For a boson that couples through $\psi^\dagger \Gamma \psi$ the amplitude, projected on the pair channel X and normalized, is $-g_{VS}^2\, F_X/(q^2 + M^2)$ with $\vec{q} = \vec{p}\,' - \vec{p}$, M the mass of the boson, and

$$F_X(\vec{p}\,', \vec{p}) = \frac{\mathrm{Tr}\big[\Phi_X(\vec{p}\,')^T\, V_1\, \Phi_X(\vec{p})^*\, V_2^T\big]}{\sqrt{N_X(\vec{p}\,')\, N_X(\vec{p})}}, \qquad V_1 = U(\vec{p}\,')^\dagger \Gamma\, U(\vec{p}), \quad V_2 = U(-\vec{p}\,')^\dagger \Gamma\, U(-\vec{p}), \tag{4.2}$$

$N_X = \mathrm{Tr}[\Phi_X \Phi_X^\dagger]$. $F_X > 0$ **means attraction.** The expression does not depend on the choice of the basis U (unit test `kernel::tests::particle_spinors`). The static approximation is used (the energy transfer is neglected in the boson propagator), as in the theory of superconductivity at weak coupling.

For the radial mode $\Gamma = \beta$ and the amplitude is called F_S; for the phase mode $\Gamma = i\beta\chi$ and it is called F_P.

4.2 Results

The program `spacetimes channels` evaluates (4.2) numerically from the Dirac matrices for both spacetimes, for the eight vector pair operators $C^-\gamma^A$ and for representative two-form pair operators $C^+\gamma^A\gamma^B$, at three momenta and thirteen scattering angles (`data/pair_channel_kernel.csv`, 1170 rows). Figure 1 shows the amplitudes for $|\vec{p}| = |\vec{p}\,'| = m$; the table gives three angles.

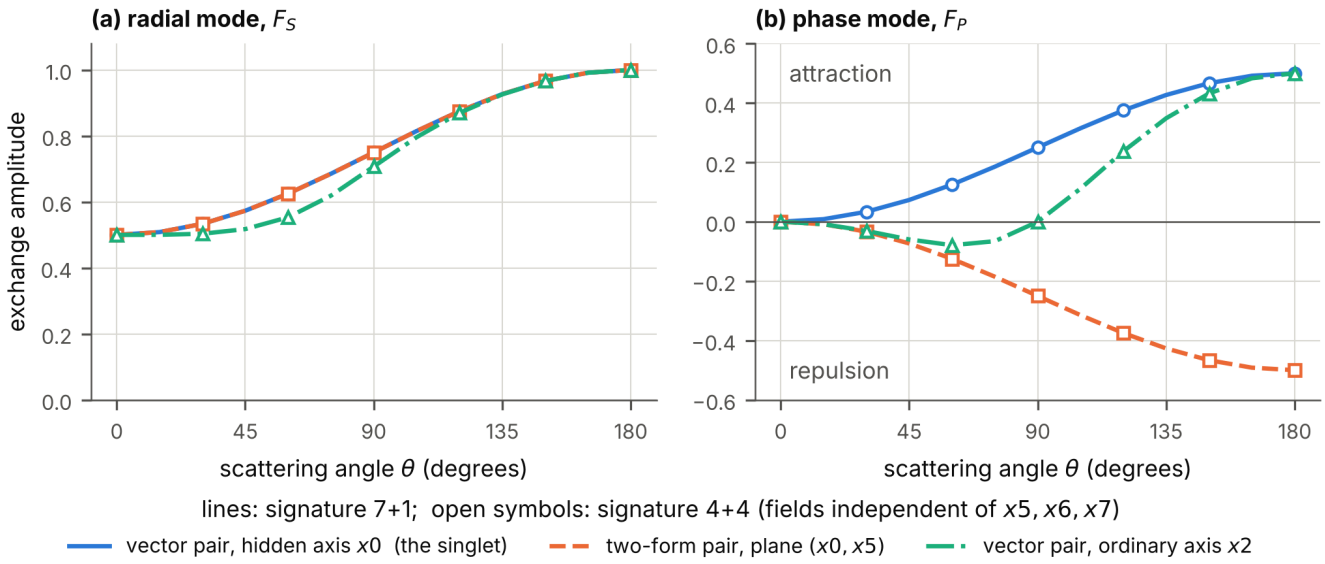


Figure 1: Exchange amplitudes of the two modes of the Vorton-Scalar in three pair channels, for momenta of magnitude m in the plane $(x1, x2)$, as functions of the scattering angle. Lines: 7+1; open symbols: 4+4. The radial mode attracts in all channels; the phase mode attracts in the vector channel along the hidden axis, repels in the two-form channel, and has both signs for a vector pair polarized in the plane of the momenta. Data: `data/pair_channel_kernel.csv`; program: `spacetimes channels`; drawn by `make_figures.py kernels`.

pair channel (7+1)	F_S, 60°	F_S, 120°	F_S, 180°	F_P, 60°	F_P, 120°	F_P, 180°
vector, hidden axis $x0$	+0.6250	+0.8750	+1.0000	+0.1250	+0.3750	+0.5000
vector, extra axis $x5$	+0.6250	+0.8750	+1.0000	+0.1250	+0.3750	+0.5000
two-form, plane $(x0, x5)$	+0.6250	+0.8750	+1.0000	-0.1250	-0.3750	-0.5000
two-form, plane $(x5, x6)$	+0.6250	+0.8750	+1.0000	-0.1250	-0.3750	-0.5000
vector, ordinary axis $x2$	+0.5534	+0.8696	+1.0000	-0.0791	+0.2372	+0.5000

Momenta of magnitude m in the plane $(x1, x2)$. The same quantities for 4+4 (fields independent of $x5, x6, x7$) differ from those for 7+1 by at most 3.0×10^{-16} over all 585 entries of `data/pair_channel_kernel.csv`.

The results, all verified by the program and identical for the two spacetimes (the largest difference between the two is given in the last line of the table):

(i) Closed forms for the hidden-axis vector pair ($X = C^{-}\gamma^{0}$). With $E = \sqrt{m^2 + p^2}$, $E' = \sqrt{m^2 + p'^2}$:

$$F_S = \frac{m^2(E+E')^2 + E^2 p'^2 + E'^2 p^2 - 2EE'\,\vec{p}\cdot\vec{p}\,'}{4E^2E'^2},$$
$$F_P = \frac{(m^2 - EE')^2 + 2(m^2 - EE')\,\vec{p}\cdot\vec{p}\,' + m^2q^2 + p^2p'^2}{4E^2E'^2}. \tag{4.3}$$

For equal energies these reduce to

$$\boxed{F_S = \frac{4m^2+q^2}{4E^2}, \qquad F_P = \frac{q^2}{4E^2} \qquad (X = C^{-}_{1671}\gamma^{0}_{1671} \text{ and } X = C^{-}_{1644}\gamma^{0}_{1644})} \tag{4.4}$$

(unit test `kernel::tests::hidden_axis_closed_forms` compares (4.3) with the matrices for 200 random momenta, on and off the energy shell; the command `channels` checks (4.4) at every tabulated point to 10^{-9}).

(ii) The radial mode does not distinguish the pair channels in which the pair has no component along the momenta: F_S is the same for the vector pairs along the four extra axes and for the two-form pairs.

(iii) The phase mode does. For the vector pairs along the four extra axes the amplitude is $F_P = +q^2/4E^2$ (equal energies): attraction at all angles. For the two-form pairs of the extra axes it is $F_P = -q^2/4E^2$: **repulsion** at all angles. For a vector pair polarized along an axis of ordinary space the amplitude depends on the directions of the momenta relative to that axis and takes both signs (the third curve of Fig. 1); such a pair has ordinary spin 1 and is not a candidate for the state sought here.

(iv) A contact attraction of Fermi type. The phase mode is massless, so its contribution to the pairing interaction of the hidden-axis pair is $-g_{VS}^2 F_P/q^2 = -g_{VS}^2/4E^2$, independent of the momentum transfer:

$$\boxed{V_{\rm NC} = -\frac{g_{VS}^2}{4E^2} = -\frac{1}{4v_{VS}^2}\,\frac{m^2}{E^2} = -\frac{G_{VS}}{2\sqrt{2}}\,\frac{m^2}{E^2}} \tag{4.5}$$

The neutral current (1.7) thus produces, between two fermions in the hidden-axis singlet channel, a point-like attraction whose strength is set by G_{VS} alone — the direct analogue of Fermi's contact interaction — and an equally strong repulsion in the two-form channels.

(v) Low-energy limit. For momenta small compared with m, $F_S \to 1$: the radial mode gives the attractive Yukawa potential

$$V_s(r) = -\alpha_{VS}\,\frac{e^{-m_s r}}{r}, \qquad \alpha_{VS} = \frac{g_{VS}^2}{4\pi}, \tag{4.6}$$

between any two fermions, whatever their internal states.

Together, (ii)–(iv) are the dynamical reason for the state proposed in the task: the radial mode binds, and the neutral current selects, among the 28 possible s-wave pairs (DM, Sec. 3.2), the four vector pairs along the extra axes — which are exactly the pairs of ordinary spin 0 that form a 4-vector — and penalizes the six spin-0 two-form pairs.

What Sec. 5 needs: the amplitudes (4.3).

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5. The gap equation

5.1 Derivation from the effective action of Cornwall, Jackiw and Tomboulis

At a fermion density n, with a chemical potential μ for the conserved fermion number, the attraction of Sec. 4 makes the Fermi sea unstable against the formation of pairs. The standard tool for such a state is the effective action for composite operators of Cornwall, Jackiw and Tomboulis (CJT), a functional $\Gamma[S]$ of the full fermion propagator S whose stationary point is the true propagator. In the doubled (Nambu–Gor'kov) description, $\Psi = (\psi, \psi^{\dagger T})$, the propagator has off-diagonal blocks that describe pair correlations, and

$$\Gamma[S] = -\frac{i}{2}\,\mathrm{Tr}\ln S^{-1} - \frac{i}{2}\,\mathrm{Tr}\big(S_0^{-1}S - 1\big) + \Gamma_2[S], \tag{5.1}$$

where S_0 is the free propagator, Γ_2 is the sum of the two-particle irreducible vacuum diagrams, and the factors $\frac{1}{2}$ compensate the doubling of the field. (Supplement S4 derives the method, states this action for the present models with all its variables, and uses it for independent checks of the results

of this section and for further predictions.) We keep in Γ_2 the diagram with one exchanged boson (radial mode plus phase mode), and make three further approximations, each of them standard and each stated here because the results depend on them:

1. *static exchange*: the boson propagators are evaluated at zero energy transfer;
2. *no change of the normal state*: the diagonal part of the self-energy is absorbed into m and μ;
3. *particles only*: the pairing of antiparticle states is neglected, which is valid when the gap and the temperature are small compared with m.

The stationarity condition $\delta\Gamma/\delta S = 0$ then becomes a linear relation between the off-diagonal self-energy (the gap) and the pair correlation, and for a gap matrix proportional to the two-particle content of one pair operator X, $\Delta(\vec{k})\,\Phi_X(\vec{k})/\sqrt{N_X/8}$, it reduces to

$$\begin{aligned} \Delta(k) &= \frac{1}{4\pi^2}\int_0^\infty dk'\,k'^2\; W(k,k')\;\frac{\Delta(k')}{E_\Delta(k')}\;\tanh\frac{E_\Delta(k')}{2T}, \\ E_\Delta(k) &= \sqrt{\xi_k^2+\Delta(k)^2}, \qquad \xi_k = \sqrt{m^2+k^2}-\mu, \end{aligned} \tag{5.2}$$

$$W(k,k') = \frac{1}{2}\int_{-1}^{1} d\cos\theta\; g_{VS}^2\left[\frac{F_S}{q^2+m_s^2}+\sigma_X\,\frac{F_P}{q^2}\right], \qquad q^2 = k^2+k'^2-2kk'\cos\theta, \tag{5.3}$$

with F_S, F_P of Eq. (4.3) and $\sigma_X = +1$ for the vector pairs along the extra axes, $\sigma_X = -1$ for their two-form pairs. For these channels the gap matrix is *unitary* (Sec. 6): all eight internal states have the same gap, which is why a single function $\Delta(k)$ suffices. The angular integral in (5.3) is done in closed form (unit test `gap::tests::angular_average_matches_quadrature`). The chemical potential is kept at its value without pairing, $\mu = \sqrt{m^2+k_F^2}$, where k_F is the Fermi momentum, $n = 8\,k_F^3/6\pi^2$ for eight internal states; this is correct as long as the gap is small compared with the Fermi kinetic energy $E_F = \sqrt{m^2+k_F^2}-m$ (the *weak-coupling* or BCS regime).

5.2 Numerical solution, and its validation

`spacetimes gap` solves (5.2) by iteration on a momentum grid that is refined logarithmically around k_F (`gap.rs`), for 120 combinations of g_{VS}, m_s/m and k_F/m, for three kernels: the vector channel ($\sigma_X = +1$), the radial mode alone, and the two-form channel ($\sigma_X = -1$). The critical temperature T_c is the temperature at which the linearized equation has the eigenvalue 1. Two validations:

- for a kernel that is constant in a shell around the Fermi surface the program reproduces the textbook results $\Delta = 2\omega_c\,e^{-1/\lambda}$ and $T_c = 0.567\,\Delta$ (unit test `gap::tests::bcs_shell_model`);
- in the scan, the ratio $T_c/\Delta(k_F)$ lies within 0.03 of 0.567 wherever the coupling is weak (checked by the command; column `Tc_over_gap_V` of `data/gap_scan.csv`).

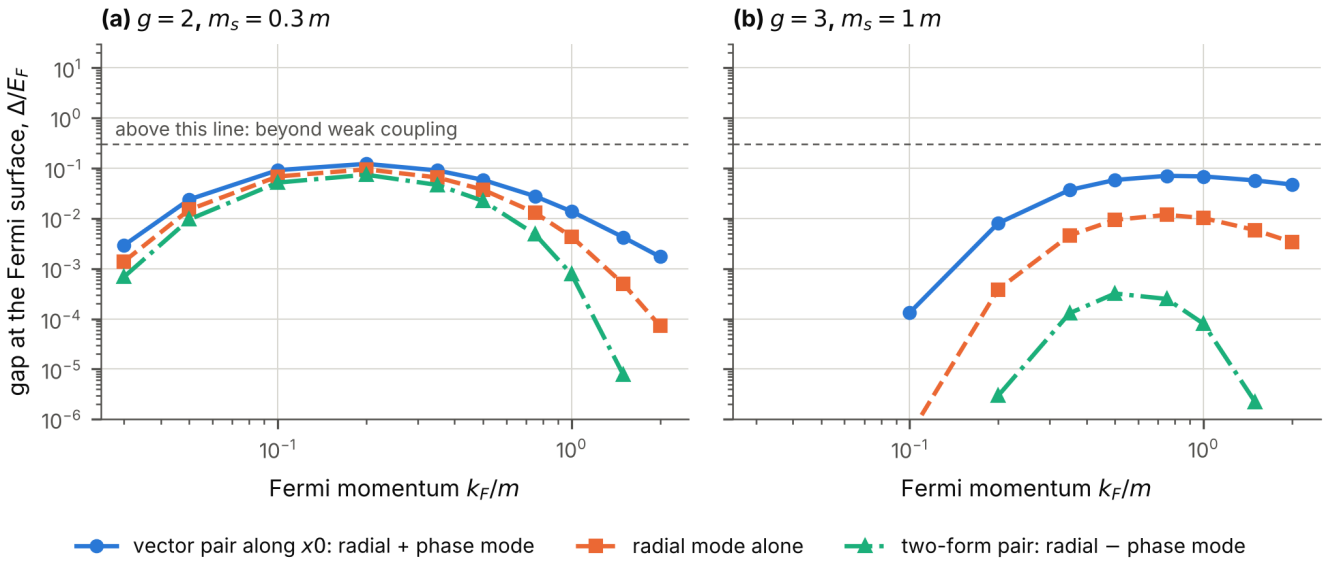


Figure 2: Gap at the Fermi surface in units of the Fermi kinetic energy, as a function of the Fermi momentum, for two sets of couplings. In every case the vector pair along an extra axis, to which the radial mode and the phase mode both contribute attraction, has the largest gap; the two-form pair, in which the phase mode repels, has the smallest. Open symbols mark points above the dashed line, where the gap is no longer small compared with the Fermi energy and Eq. (5.2) with a fixed chemical potential is not quantitative. Data: `data/gap_scan.csv`; program: `spacetimes gap`.

g_{VS}	m_s/m	k_F/m	λ_S	λ_P	vector: Δ/E_F	radial only	two-form	T_c/Δ (vector)	coupling
1	0.1	0.1	0.204	0.0013	2.5×10^{-2}	2.4×10^{-2}	2.3×10^{-2}	0.567	weak
1	0.1	0.35	0.137	0.0042	1.4×10^{-3}	1.1×10^{-3}	8.4×10^{-4}	0.567	weak
1	0.1	1	0.063	0.0090	8.2×10^{-7}	$< 10^{-7}$	$< 10^{-7}$	0.567	weak
2	0.3	0.1	0.186	0.0050	9.1×10^{-2}	6.8×10^{-2}	5.2×10^{-2}	0.565	weak
2	0.3	0.35	0.266	0.0167	9.0×10^{-2}	6.5×10^{-2}	4.6×10^{-2}	0.565	weak
2	0.3	1	0.169	0.0358	1.4×10^{-2}	4.3×10^{-3}	7.7×10^{-4}	0.567	weak
3	1	0.1	0.045	0.0113	1.3×10^{-4}	4.9×10^{-7}	$< 10^{-7}$	0.567	weak
3	1	0.35	0.130	0.0377	3.7×10^{-2}	4.6×10^{-3}	1.3×10^{-4}	0.567	weak
3	1	1	0.178	0.0806	6.9×10^{-2}	1.0×10^{-2}	8.0×10^{-5}	0.566	weak

λ_S and λ_P are the dimensionless couplings $N(0)W(k_F, k_F)$ of the radial and of the phase mode. "$< 10^{-7}$" means below the resolution of the calculation. Source: `data/gap_scan.csv` (all 120 combinations).

The table and Fig. 2 show the result that matters for the task: **the gap of the vector channel exceeds that of the two-form channel**, by a factor that grows with the density and with the mass of the radial mode — from a factor of a few to several orders of magnitude. The phase mode alone (the neutral current without the radial mode) gives a gap that is negligible at low density, because its dimensionless coupling $\lambda_P = N(0)\, g_{VS}^2/4E_F'^2$ (with $N(0) = k_F E_F'/2\pi^2$ the density of states per internal state and $E_F' = \sqrt{m^2 + k_F^2}$) vanishes with k_F; it becomes comparable to the radial contribution when $k_F \sim m$.

5.3 From overlapping pairs to bound pairs

At low density the attraction (4.6) is felt by two fermions alone. They form a bound state if and only if (`spacetimes dimer`; `data/dimer.csv`; Fig. 3)

$$\boxed{D_{VS} \equiv \frac{\alpha_{VS}\, m}{m_s} > D_c = 1.67981} \tag{5.4}$$

The program integrates the radial Schrödinger equation $u'' + D_{VS}\, e^{-\rho} u/\rho = \varepsilon u$ ($\rho = m_s r$, $\varepsilon = mE_b/m_s^2$, E_b the binding energy); the critical value agrees with the value 1.6798 known for the Yukawa potential, the scattering length has the Born limit $a = -D_{VS}/m_s$ for weak coupling, and the binding energy approaches the screened-Coulomb expansion $\varepsilon = D_{VS}^2/4 - D_{VS} + 3/2 - 2/D_{VS}$ for strong coupling (unit tests of `dimer.rs`).

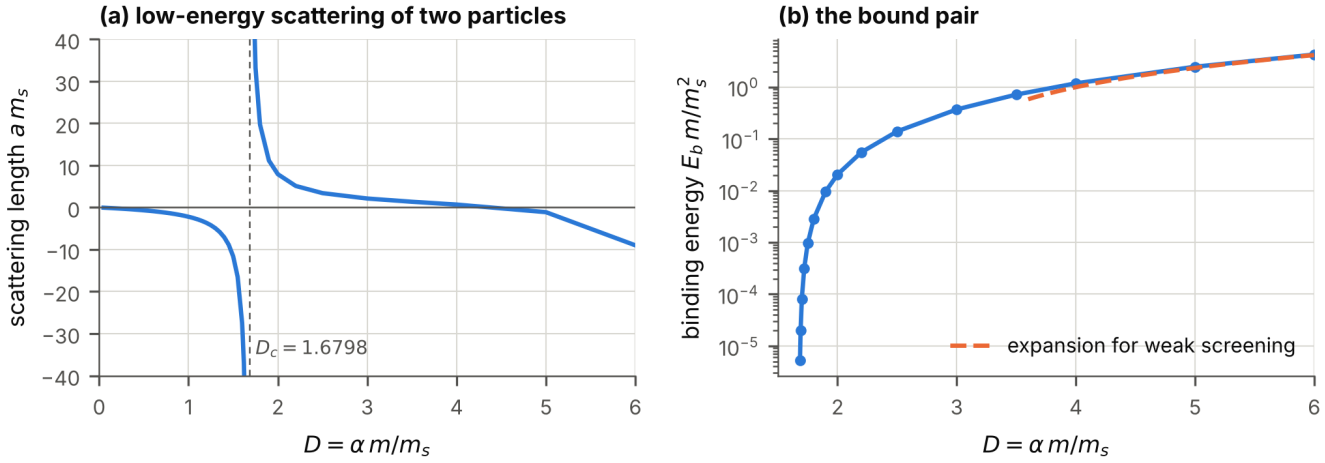


Figure 3: (a) Scattering length of two fermions in the Yukawa potential of the radial mode, in units of $1/m_s$; it diverges at the critical coupling, where the bound state appears. (b) Binding energy of the pair. Data: `data/dimer.csv`; program: `spacetimes dimer`.

For $D_{VS} < D_c$ the superfluid is of the BCS kind at all densities: large, overlapping pairs, described by Eq. (5.2). For $D_{VS} > D_c$ a dilute gas consists of bound pairs — bosons of spin 0 — that condense; as the density rises the pairs overlap and the state crosses over continuously to the BCS kind. In the scan of Sec. 5.2 the points marked “beyond weak coupling” are on the way to this regime; there the *existence* of the paired state is not in doubt (two fermions already bind), but the numbers of Eq. (5.2) are not quantitative. With eight internal states and an attraction between all of them, bound states of more than two fermions are to be expected as well; they are discussed in Supplement S3, Sec. 4.

What Sec. 6 needs: the statement that the four vector pairs along the extra axes have the strongest pairing.

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6. The order parameter and the ground state

6.1 The gap matrix

The condensate is described by the expectation values of the four vector pair operators along the extra axes,

$$\Delta^a \propto \langle \psi^T C^- \gamma^a \psi \rangle, \qquad a \in \{0, 5, 6, 7\},$$

four complex numbers. On the eight internal states of the pair $(\vec{p},-\vec{p})$ they define the **gap matrix**

$$\mathsf{M}(\vec{p})=\sum_a \Delta^a\,\frac{\Phi_a(\vec{p})}{\sqrt{N_a/8}},\qquad \Phi_a=U(\vec{p})^T C^-\gamma^a U(-\vec{p}), \tag{6.1}$$

and the energies of the quasi-particles are $\sqrt{\xi^2+d_r^2}$, where d_r^2 ($r=1,\dots,8$) are the eigenvalues of $\mathsf{M}\mathsf{M}^\dagger$.

6.2 The Ginzburg–Landau energy

Near the critical temperature the free energy can be expanded in the order parameter (Ginzburg and Landau). At weak coupling the expansion is

$$\mathcal{F}=a(T)\,\mathrm{Tr}\big[\mathsf{M}\mathsf{M}^\dagger\big]+\frac{b}{2}\,\mathrm{Tr}\big[(\mathsf{M}\mathsf{M}^\dagger)^2\big]+K\,\mathrm{Tr}\big[\nabla\mathsf{M}\cdot\nabla\mathsf{M}^\dagger\big]+\dots,$$
$$a(T)=\frac{N(0)}{2}\ln\frac{T}{T_c},\qquad b=\frac{7\zeta(3)}{16\pi^2T_c^2}N(0)>0,$$

with $N(0)$ the density of states at the Fermi surface per internal state (for a unitary state $\mathsf{M}\mathsf{M}^\dagger=|\Delta|^2$ this is the textbook energy of four independent copies of a spin-$\frac12$ superconductor, with the minimum $|\Delta|^2=\frac{8\pi^2}{7\zeta(3)}T_c^2(1-T/T_c)$), so the form of the quartic term is decided by one matrix identity. The program verifies it for both spacetimes (`spacetimes channels`; unit test `kernel::tests::ginzburg_landau_quartic_invariant`; `data/gl_quartic_check.csv`; largest deviation 2.1×10^{-14} for 7+1 and 1.8×10^{-14} for 4+4):

$$\boxed{\frac18\mathrm{Tr}\big[\mathsf{M}\mathsf{M}^\dagger\big]=\sum_a|\Delta^a|^2,\qquad \frac18\mathrm{Tr}\big[(\mathsf{M}\mathsf{M}^\dagger)^2\big]=2\Big(\sum_a|\Delta^a|^2\Big)^2-\big|\eta_{ab}\,\Delta^a\Delta^b\big|^2} \tag{6.2}$$

where η_{ab} is the metric of the spacetime restricted to the four extra axes: $\eta_{1671}=\mathrm{diag}(-1,-1,-1,-1)$ on them (all four are space-like), and $\eta_{1644}=\mathrm{diag}(+1,-1,-1,-1)$ (the hidden axis is space-like, the other three are time-like). The signature of the spacetime thus enters the energy of the superfluid through the quadratic form $\eta_{ab}\Delta^a\Delta^b$.

Origin of (6.2) in 7+1. For particles at rest $\Phi_a=-\tau^a_{1671}$ (check `71-09`), real antisymmetric 8×8 matrices that anticommute and square to -1. Then $\mathsf{M}\mathsf{M}^\dagger=\sum_a|\Delta^a|^2-\sum_{a<b}(\Delta^a\Delta^{b*}-\Delta^b\Delta^{a*})\tau^a\tau^b$, and since the products $\tau^a\tau^b$ are traceless and mutually trace-orthogonal with $(\tau^a\tau^b)^2=-1$, the trace of the square is $8(\sum|\Delta^a|^2)^2+8\sum_{a<b}|\Delta^a\Delta^{b*}-\Delta^b\Delta^{a*}|^2$, which equals $8[2(\sum|\Delta|^2)^2-|\sum_a(\Delta^a)^2|^2]$.

6.3 The ground state

At fixed $\sum_a|\Delta^a|^2$ the quartic term (6.2) is smallest when $|\eta_{ab}\Delta^a\Delta^b|$ is largest, i.e. equal to $\sum_a|\Delta^a|^2$. Then $\mathsf{M}\mathsf{M}^\dagger=\sum_a|\Delta^a|^2\cdot1$: the state is **unitary**, all eight quasi-particle branches have the same gap, and the condensation energy is the largest possible. The minimizing configurations are

- in 7+1: $\Delta^a=|\Delta|\,e^{i\varphi}\,n^a$ with a *real* unit vector n^a in the space of the four extra axes;
- in 4+4: $\Delta^0=|\Delta|e^{i\varphi}\,n^0$, $\Delta^t=\pm i\,|\Delta|e^{i\varphi}\,n^t$ with a real unit vector (n^0,n^5,n^6,n^7): the components along the time-like axes carry a relative phase $\pm i$.

The **hidden-axis singlet** of the task,

$$\boxed{\begin{array}{l}\Delta^a=\Delta\,\delta^{a0}:\\ \langle\psi^T_{1671}C^-_{1671}\gamma^0_{1671}\psi_{1671}\rangle\neq0\quad\text{(7+1)},\qquad \langle\psi^T_{1644}C^-_{1644}\gamma^0_{1644}\psi_{1644}\rangle\neq0\quad\text{(4+4)},\end{array}} \tag{6.3}$$

is a member of this family in both spacetimes. Its properties, proved in "dirac matrices" (Sec. 3.3): the pair has ordinary spin 0; it is invariant under the rotations of ordinary space and of the coordinates $x5, x6, x7$; and it is invariant under the Lorentz transformations of the seven coordinates other than $x0$ — in particular it is a *scalar* in ordinary spacetime, although it is a component of a vector in eight dimensions.

How the hidden axis is singled out. By Sec. 2.1 the order parameter is a 4-vector of the internal $SO(4)$ of the zero-mode sector in both spacetimes; in 4+4 the components that form the real 4-vector are $(\Delta^0, i\Delta^5, i\Delta^6, i\Delta^7)$, which is why the relative phase $\pm i$ appears above. All directions of this 4-vector have exactly the same energy.

- *In 7+1* the direction n^a is chosen spontaneously, and calling the chosen one "the hidden axis" is a choice of coordinates. The symmetry breaking is $SO(4) \to SO(3)$ together with $U(1)_F \to Z_2$; besides the sound mode of the superfluid (the Goldstone mode of $U(1)_F$) there are three Goldstone modes, the rotations of n^a. Because the $SO(4)$ is a symmetry of the full theory, nothing within the 7+1 model prefers one direction.
- *In 4+4* the same holds within the zero-mode sector: the singlet Δ^0 of the rotations $SO(3)_T$ of the additional times and their triplet Δ^t are degenerate **exactly**, to all orders, because the hidden $SO(4)$ of Sec. 2.1 rotates one into the other; the same four gapless modes exist. But the hidden symmetry is a property of the zero-mode sector only. Every effect that involves a dependence of the fields on $x0$ — in Stage 2, excitations that travel around the hidden ring, with energies of at least the inverse radius — breaks it down to $SO(3)_T$, under which the hidden axis *is* distinguished. Such effects lift the degeneracy between singlet and triplet and give the three additional modes a mass. Which of the two then lies lower has not been determined; that it is the singlet (6.3) is the hypothesis of the task.

(An earlier version of this supplement called the degeneracy in 4+4 accidental. That was wrong; the symmetry analysis of Sec. 2.1, made in the course of the checks of Supplement S4, corrected it. Error log, E26.)

In Stage 2 of the author's programme the hidden coordinate is curled up and the other three coordinates are treated differently; this breaks the $SO(4)$ of 7+1 explicitly as well, and would decide the direction in both spacetimes. It is not implemented here.

6.4 Collective modes and the sound of the superfluid

The phase φ of the order parameter is the Goldstone field of the broken fermion number. Its excitation is a density wave, the first sound of the superfluid (the Bogoliubov–Anderson mode), with the velocity given by the compressibility, $c_s^2 = dP/d\rho$. In the weak-coupling regime the pressure is the degeneracy pressure of the fermions, and

$$c_s^2 = \frac{k_F^2}{3\,(m^2 + k_F^2)} = \frac{v_F^2}{3} \quad \text{(weak coupling)}; \qquad c_s^2 = \frac{4\pi a_{BB}\, n_B}{m_B^2} \quad \text{(dilute gas of bound pairs)}, \tag{6.4}$$

where v_F is the Fermi velocity and, in the second expression (Bogoliubov), m_B, n_B and $a_{BB} > 0$ are the mass, the density and the mutual scattering length of the bound pairs. Equation (6.4) is the input of the cosmological simulations of Supplement S3, in which the superfluid appears as a fluid with this sound speed.

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7. The superfluid field equations (summary)

For the reader who wants the complete set of equations in one place. *Normal phase, 7+1*: Eqs. (1.8); *normal phase, 4+4 (quantized sector)*: Eqs. (3.2). *Paired phase*, both spacetimes, in the mean-field approximation of Sec. 5: the fermion field obeys the Bogoliubov–de Gennes equations

$$i\partial_4 \psi = (\mathsf{h} - \mu)\,\psi + \hat{\Delta}\,\psi^{\dagger T}, \qquad \hat{\Delta}(\vec{p}) = U(\vec{p})^*\,\mathsf{M}(\vec{p})\,U(-\vec{p})^\dagger, \tag{7.1}$$

in which $\hat{\Delta}$ is the 16×16 pair potential built from the gap matrix (6.1), and the gap obeys the self-consistency equation (5.2) with the kernel (5.3). Near T_c the order parameter varies slowly and obeys the Ginzburg–Landau equation that follows from Sec. 6.2,

$$-K\,\nabla^2\Delta^a + a(T)\,\Delta^a + b\Big[2\Big(\sum_c |\Delta^c|^2\Big)\Delta^a - \big(\eta_{cd}\Delta^c\Delta^d\big)\,\eta_{ab}\,\Delta^{b*}\Big] = 0, \qquad K = \frac{7\zeta(3)}{96\pi^2 T_c^2} N(0)\,v_F^2, \tag{7.2}$$

obtained by varying the energy of Sec. 6.2 with respect to Δ^{a*} and using (6.2). For a uniform unitary state it gives $a + b|\Delta|^2 = 0$, the minimum quoted in Sec. 6.2. The stiffness K is the standard weak-coupling value for an isotropic unitary state (it corresponds to the coherence length $\xi^2 = K/|a| = 7\zeta(3)v_F^2/[48\pi^2 T_c^2(1-T/T_c)]$); it has not been re-derived for this model and is quoted for orientation only.

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8. Vortices and vortons (what the name refers to)

This section collects known results that apply to the Vorton-Scalar; none of them is computed in this project, and each is attributed. Two kinds of vortex must be kept apart.

Vortices of the superfluid. The order parameter Δ^a has a phase, the Goldstone field of fermion number. A line around which this phase winds by 2π is a quantized vortex of the superfluid, with circulation $2\pi/(2m)$ per unit winding (Onsager, Feynman; the pair has mass $2m$). In its core the condensate vanishes and the scalar density of the fermions differs from its value in the bulk; by the second of Eqs. (1.8) so does the Vorton-Scalar. Every such vortex therefore carries a tube of displaced Vorton-Scalar field along with it. This is the sense in which the scalar was named in the NCSS project.

Vortices of the Vorton-Scalar itself. A complex scalar with the potential of (1.1) has static solutions in which the phase winds by $2\pi n$ around a line and $|\Phi_{VS}|$ vanishes on the line (Nielsen and Olesen for the gauged case; for a global symmetry such as $U(1)_{VS}$ the energy per length grows logarithmically with the distance).

Zero modes. A fermion that obtains its mass from the scalar through a Yukawa coupling of the form (1.1) is massless on the line, and the Dirac equation in the background of the vortex has $|n|$ normalizable solutions of zero energy that move along the line at the speed of light (Jackiw and Rossi). The first of Eqs. (1.8) is of exactly this type.

Currents and vortons. Zero modes make the string a perfect conductor of the corresponding charge (Witten's superconducting strings; here the charge is fermion number). A closed loop carrying such a current has a conserved number of charge carriers and a conserved winding of the current; the resulting angular momentum can balance the tension of the string and prevent the loop from collapsing. Such stationary loops are **vortons** (Davis and Shellard). Whether they are stable, and with what abundance they would be produced, depends on details that have been studied for other models and are open here. The NCSS project used a real scalar, which has no such defects, and its documents state that its Vorton-Scalar is unrelated to these vortons; for the complex field of the present project the relation is literal, but nothing below depends on it.

The hidden ring. The author's hypothesis for Stage 2 is that $x0$ curls up into a ring. A Vorton-Scalar whose phase winds *around that ring*, $\Phi_{VS} \propto e^{i\,n\,x0/R}$, is a configuration with a constant neutral current $J^0_{VS} = n\,v_{VS}^2/R$ along the hidden axis. By (1.7) it acts on the fermions as a constant axial potential along $x0$. This is a natural object to study in Stage 2; nothing about it is claimed here.

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9. What has been established, and what has not

Established (each item with its evidence):

1. A field with the required properties exists and is defined by the Lagrangian (1.1) in 7+1: a complex scalar whose neutral current (1.6) couples to the axial current of the fermions with a Fermi-type constant (1.7), and from which the fermion mass originates (1.5). (Algebra of Sec. 1; the matrix properties used are machine-checked.)
2. In 4+4 the same construction is impossible in the fully covariant theory (Sec. 3.1) and possible in the quantized sector of fields independent of the additional times (Sec. 3.2).
3. One-boson exchange is attractive in the hidden-axis vector channel for both modes, with the closed forms (4.4); the neutral current gives a contact attraction (4.5) in that channel and repulsion in the two-form channels. (Numerical evaluation from the matrices, both spacetimes; closed forms verified.)
4. The mean-field gap equation has solutions; the vector channel has the largest gap; the numerical method reproduces the textbook limits. (Sec. 5.)
5. Two fermions bind when $D_{VS} > 1.6798$. (Sec. 5.3.)
6. The quartic Ginzburg–Landau invariant is (6.2), with the metric of the spacetime; the hidden-axis singlet (6.3) is among the states of lowest energy. (Machine check, both spacetimes.)
7. In the sector of fields independent of the extra coordinates both models have the internal symmetry $SU(2)_A \times SU(2)_B \times U(1)_F \times U(1)_E$, under which the order parameter is a 4-vector; in 4+4 this symmetry is hidden (Sec. 2.1; machine check).

Not established:

1. That the hidden axis is *preferred*: in 7+1 the direction is spontaneous; in 4+4 singlet and triplet are exactly degenerate in the zero-mode sector, and which is lower once the fields depend on $x0$ has not been computed (Sec. 6.3).
2. Anything beyond the mean-field, static, particles-only approximation: retardation, the renormalization of the normal state, fluctuations of the order parameter, and the strong-coupling regime beyond the open symbols of Fig. 2.
3. The consistency of the 4+4 model with the full spacetime (the restriction to fields independent of $x5, x6, x7$ is an assumption).
4. The high-energy completion of the theory (the couplings have negative mass dimension in eight dimensions).
5. Stability and abundance of vortons; anything about Stage 2.
6. Whether nature contains this field. The consequences for the Higgs hypothesis are examined in Supplement S2, those for cosmology in Supplement S3; Supplement S4 re-derives the results of Secs. 5 and 6 from the effective action of Cornwall, Jackiw and Tomboulis and adds what that action predicts for the unpaired state, for finite temperature and for the vacuum.

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10. Provenance of the numbers and figures

Item	Produced by	File
amplitudes F_S, F_P (Fig. 1, table of Sec. 4.2)	`spacetimes channels`	`data/pair_channel_kernel.csv`
closed forms (4.3), (4.4)	unit test `kernel::tests::hidden_axis_closed_forms`; `spacetimes channels`	—
gaps and critical temperatures (Fig. 2, table of Sec. 5.2)	`spacetimes gap`	`data/gap_scan.csv`, `data/gap_profile.csv`
critical coupling, scattering length, binding energy (Fig. 3)	`spacetimes dimer`	`data/dimer.csv`
identity (6.2)	`spacetimes channels`; unit test `kernel::tests::ginzburg_landau_quartic_invariant`	`data/gl_quartic_check.csv`

Item	Produced by	File
internal symmetry (Sec. 2.1)	`spacetimes symmetry`; unit tests of `symmetry.rs`	`data/reduced_symmetry.csv`
figures	`python3 code/scripts/make_figures.py kernels gap dimer`	`figures/fig_kernels`, `fig_gap`, `fig_dimer` (.pdf, .png)
tables of this document	`python3 code/scripts/make_tables.py`	`provenance/generated/tables.md`

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